

Multiagent Optimization for Coordinated Operations of Transmission and Distribution System

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Abstract

Distributed energy resources are transforming distribution systems (DS) from passive load-serving networks into active participants in grid operations and electricity markets. This creates a need for coordinated transmission-distribution system operations while preserving the decision-making autonomy and protecting the operational information of individual DSs. In this paper, we propose a multiagent optimization framework in which multiple DSs interact with the transmission system at shared market nodes. Each DS solves its own SOCP-relaxed optimal power flow problem locally to determine the operational decisions. It shares only transaction decisions with the markets, and market prices are updated using aggregated transactions incurred at the nodes. The resulting model is a standard noncooperative Nash game, where coupling among DSs arises only through market interactions. To compute equilibrium solutions, we develop three proximal best-response algorithms with different update rules: cyclic coordinated, cyclic immediate, and random immediate updates. To accommodate uncertain electricity demand, we further present a two-stage stochastic programming setting and tackle the problem using sample average approximation-based proximal best-response schemes. Our numerical experiments on three-DS and six-DS systems show that immediate-update schemes outperform coordinated updates, especially in larger systems. The stochastic results and out-of-sample validation further demonstrate that SAA improves the robustness of first-stage decisions under demand uncertainty.

History: First version: August, 2026; Current version: September 3, 2026.

1 Introduction

The increasing penetration of distributed energy resources (DERs), including renewable generation, flexible loads, and energy storage, has transformed the role of distribution systems (DSs) in power system operations and electricity markets [International Energy Agency, 2022, Adham et al., 2025, Lee, 2026]. Traditionally, DSs were treated as passive load-serving networks whose internal loads were aggregated and passed to the transmission system (TS) as equivalent

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demand. However, under high DER penetration, DSs can actively adjust generation, demand, and power exchanges with the grid. Consequently, operational interaction between the TS and DSs has become more important for maintaining reliability, improving efficiency, and enabling DER market participation.

Recent studies have developed integrated transmission-distribution (T-D) models to capture the coupling between the TS and multiple DSs [Ospina et al., 2023]. In this line of work, researchers have proposed coordinated T-D optimal power flow (OPF) models that explicitly represent physical interactions at the T-D interfaces [Li et al., 2016]. To improve computational scalability and attain organizational separation between T-D system operators, subsequent studies have further developed decentralized OPF methods for integrated T-D grids [Lin et al., 2019], hierarchical and decomposition-based coordination models [Liu et al., 2020], and distributed OPF formulations for generic integrated T-D systems [Dai et al., 2025]. These studies show that coordinated T-D models more accurately represent modern power system operations than approaches that treat TS and DSs independently. However, most existing T-D coordination models are formulated from a cooperative or system-wide optimization perspective, where a common objective is optimized through centralized control or coordinated information exchange among system operators.

In practice, DSs may be operated by different entities and therefore be unwilling to share detailed private information, such as network topology, generation costs, and local operating constraints. Moreover, each DS may make independent operational and market decisions based on its own objective, rather than following a centralized system-level objective. These considerations motivate a multiagent formulation in which DSs interact with the TS through shared market nodes rather than direct physical or informational connections. In this setting, each DS independently determines its electricity transactions, while market prices reflect the aggregate decisions of all participating DSs. Such interactions naturally lead to a noncooperative game, where each DS optimizes its own decision while taking the decisions of other DSs as given.

Distributed algorithms provide a natural computational framework for multiagent models. Classical distributed optimization methods, including dual decomposition, alternating direction method of multipliers (ADMM), and block coordinate descent, have been widely used to decompose large-scale optimization problems [Bertsekas and Tsitsiklis, 1989, Boyd et al., 2011, Razaviyayn et al., 2013, Wright, 2015]. In power systems, these methods have further motivated coordination algorithms for distributed OPF [Lin et al., 2019, Dai et al., 2025, Erseghe, 2014, Peng and Low, 2016]. While these methods work well for cooperative optimization problems with a common system-level objective, multiagent market interactions are often better modeled as noncooperative games, in which each agent optimizes its own objective. For such problems, best-response (BR) schemes provide a natural approach, where each player sequentially solves its own optimization problem while taking the other players' strategies as given [Fudenberg and Levine, 1998, Nisan et al., 2011]. However, standard BR schemes may fail to converge when the coupling among players is strong or when the best-response mapping is not contractive. To improve stability, proximal regularization can be incorporated into each player's objective, yielding proximal BR algorithms that promote more stable convergence behavior [Facchinei et al., 2011]. Several variants of BR-type algorithms have been developed in [Scutari et al., 2014, Yi et al., 2022].

Uncertainty further complicates T-D coordination. In particular, net-demand in DSs is uncertain due to load variability, renewable generation, and forecasting errors. This motivates a two-stage stochastic formulation in which first-stage market transaction decisions are made before uncertainty is realized, while second-stage dispatch and network decisions are adjusted after demand realization. Sample average approximation (SAA) provides a tractable approach for

approximating the expected second-stage cost by a finite set of scenarios [Kleywegt et al., 2002]. Combining SAA with proximal BR schemes provides a possible approach to compute stochastic equilibrium decisions.

In this paper, we propose a multiagent framework for coordinated T-D system operations with multiple independent DSs. The DSs interact with the TS through shared market nodes, where aggregate transactions endogenously determine prices through inverse demand functions. Each DS solves a local SOCP-relaxed distribution-system optimization problem and only shares its transaction decisions with the markets. This leads to a standard noncooperative Nash game in which DSs are coupled through shared market prices.

The main contributions of this paper are summarized as follows.

- We propose a market-based multiagent T-D coordination model in which multiple DSs make independent electricity buying and selling decisions through shared market nodes, with prices determined by inverse demand functions and local DS feasibility represented by SOCP-relaxed network constraints.
- We formulate the deterministic problem as a noncooperative Nash game and develop three proximal best-response algorithms, including cyclic coordinated, cyclic immediate, and random immediate update schemes.
- We extend the framework to a stochastic setting with uncertain electricity demand and combine the proximal BR schemes with SAA. This allows the first-stage transaction decisions to account for demand uncertainty while maintaining decentralized decision-making.
- We validate the proposed framework on three-DS and six-DS market settings and compare convergence behavior, transaction decisions, market prices, and computational performance under deterministic and stochastic cases.

The remainder of this paper is organized as follows. In Section 2, we present the market setting, DS optimization model, and game-theoretic formulation. Section 3 develops the deterministic and stochastic proximal BR algorithms. Section 4 reports the numerical experiments and validation results. Finally, in Section 5, we share the conclusions of our work and identify directions for future research.

2 Market Setting and Models

We consider a system consisting of multiple geographically separated DSs, indexed by the set $\mathcal{N}$. Each DS interacts with TS through a set of interconnection lines, denoted by $\mathcal{I}$, which are associated with transmission-side market nodes indexed by $\mathcal{M}$. These market nodes act as coupling interfaces between DSs and the TS, through which energy transactions take place. Fig. 1 illustrates the considered market structure. We assume no direct connections among the DSs; consequently, they cannot exchange electricity or share information with one another.

2.1 DS Optimization Problem Elements

For each DS $n \in \mathcal{N}$, we denote the set of buses and lines by $\mathcal{B}_n$ and $\mathcal{L}_n$, respectively. The set of DERs within DS n is denoted by $\mathcal{G}_n$, where each DER represents a small-scale modular generation unit, such as photovoltaic systems, wind turbines, or combustion turbines.

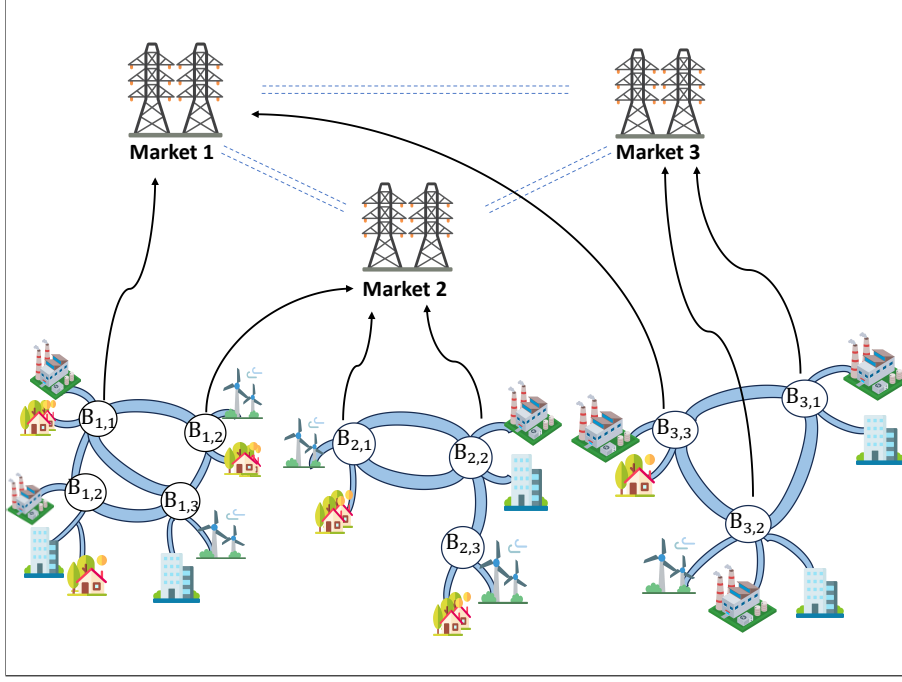


Figure 1: Market setting.

For each DER $i \in \mathcal{G}_n$, let $p_{n,i}^g$ and $q_{n,i}^g$ denote its active and reactive (where applicable) power outputs, respectively. These outputs are subject to capacity constraints, given by the bounds $[p_{n,i}^{g,\min}, p_{n,i}^{g,\max}]$ for active power and $[q_{n,i}^{g,\min}, q_{n,i}^{g,\max}]$ for reactive power. Specifically,

$$p_{n,i}^{g,\min} \leq p_{n,i}^g \leq p_{n,i}^{g,\max} \quad \forall i \in \mathcal{G}_n, \quad (1a)$$

$$q_{n,i}^{g,\min} \leq q_{n,i}^g \leq q_{n,i}^{g,\max} \quad \forall i \in \mathcal{G}_n. \quad (1b)$$

Furthermore, we assume that each bus $i \in \mathcal{B}_n$ is associated with an electricity demand. Let $p_{n,i}^d$ and $q_{n,i}^d$ denote the active and reactive power demands at bus i , respectively.

Power flows within a DS are inherently nonconvex functions of the bus voltages. Let $V_{n,i}$ denote the complex voltage at bus $i \in \mathcal{B}_n$, which we represent in rectangular form as

$$V_{n,i} = e_{n,i} + \mathbf{i}f_{n,i},$$

where $\mathbf{i} = \sqrt{-1}$. To model power flows, we use the network's nodal admittance matrix. For each line $(i, j) \in \mathcal{L}_n$, the corresponding admittance is given by

$$Y_{n,ij} = G_{n,ij} + \mathbf{i}B_{n,ij}.$$

The diagonal elements of the admittance matrix are defined as

$$G_{n,ii} = g_{n,ii} - \sum_{j \neq i} G_{n,ij}, \quad B_{n,ii} = b_{n,ii} - \sum_{j \neq i} B_{n,ij},$$

where $g_{n,ii}$ and $b_{n,ii}$ denote the shunt conductance and susceptance at bus $i \in \mathcal{B}_n$, respectively. We define $c_{n,ii} = e_{n,i}^2 + f_{n,i}^2$ to represent the squared voltage magnitude, i.e., $|V_{n,i}|^2$. Furthermore, for the voltage $V_{n,i}V_{n,j}^*$, where $(\cdot)^*$ denotes the complex conjugate, we define

$$c_{n,ij} = e_{n,i}e_{n,j} + f_{n,i}f_{n,j}, \quad s_{n,ij} = e_{n,i}f_{n,j} - e_{n,j}f_{n,i},$$

respectively. To address the computational challenges arising from the nonconvexity of power flow equations, we adopt their convex second-order cone programming (SOCP) relaxation proposed in [Kocuk et al., 2016]. The following constraints characterize the relaxation:

$$p_{n,i}^f = G_{n,ii}c_{n,ii} + \sum_{j:(i,j) \in \mathcal{L}_n} (G_{n,ij}c_{n,ij} - B_{n,ij}s_{n,ij}) \quad \forall i \in \mathcal{B}_n \quad (2a)$$

$$q_{n,i}^f = -B_{n,ii}c_{n,ii} + \sum_{j:(i,j) \in \mathcal{L}_n} (-B_{n,ij}c_{n,ij} - G_{n,ij}s_{n,ij}) \quad \forall i \in \mathcal{B}_n \quad (2b)$$

$$c_{n,ij} = c_{n,ji}, \quad s_{n,ij} = -s_{n,ji}, \quad (i, j) \in \mathcal{L}_n, \quad (2c)$$

$$c_{n,ij}^2 + s_{n,ij}^2 \leq c_{n,ii}c_{n,jj}, \quad (i, j) \in \mathcal{L}_n, \quad (2d)$$

$$(V_{n,i}^{\min})^2 \leq c_{n,ii} \leq (V_{n,i}^{\max})^2, \quad i \in \mathcal{B}_n. \quad (2e)$$

In the above, $p_{n,i}^f$ and $q_{n,i}^f$ denote the net active and reactive power injections from bus i into the distribution network.

The following equations capture the power flow balance at each bus:

$$\sum_{j \in \mathcal{G}_{n,i}} p_{n,j}^g - \sum_{j \in \mathcal{B}_{n,i}} p_{n,j}^d - p_{n,i}^f = 0, \quad \forall i \in \mathcal{B}_n, \quad (3a)$$

$$\sum_{j \in \mathcal{G}_{n,i}} q_{n,j}^g - \sum_{j \in \mathcal{B}_{n,i}} q_{n,j}^d - q_{n,i}^f = 0, \quad \forall i \in \mathcal{B}_n. \quad (3b)$$

The DS optimization model minimizes the total system cost. The generation cost of each DER $i \in \mathcal{G}_n$ is modeled using a quadratic cost function with parameters α_i and β_i . Accordingly, the total system cost is given by:

$$f_n(p_n^g) = \sum_{i \in \mathcal{G}_n} \alpha_i (p_{n,i}^g)^2 + \beta_i p_{n,i}^g \quad (4)$$

At each DS, we aim to meet demand at all buses at minimum generation cost while ensuring that power flows satisfy the underlying physical laws.

2.2 DS-Markets Interactions

We denote by $\mathcal{I}_n$ the set of interconnections linking the n -th DS to the markets. Each interconnection $(i, m) \in \mathcal{I}_n$ connects a bus $i \in \mathcal{B}_n$ to a market $m \in \mathcal{M}$. The set of all interconnections is given by $\mathcal{I} := \{(n, i, m) : n \in \mathcal{N}, (i, m) \in \mathcal{I}_n\}$.

We define y_{nim} as the transaction amount on the interconnection between bus i of DS n and market m . A positive value of y_{nim} indicates that DS n sells electricity to market m , whereas a negative value indicates a purchase. Accordingly, the total transaction amount at market $m \in \mathcal{M}$ is given by

$$s_m = \sum_{n \in \mathcal{N}} \sum_{i:(i,m) \in \mathcal{I}_n} y_{nim}. \quad (5)$$

At each market, the transaction price is determined by an inverse demand function. As illustrated in Fig. 2, the price decreases as the net supply increases ($s_m \geq 0$), and increases as

the net supply decreases ($s_m \leq 0$). We take the day-ahead locational marginal price (LMP) at the TS, denoted by μ_m , as the baseline price for market $m \in \mathcal{M}$. The parameter $\gamma_m > 0$ represents the sensitivity of the price to variations in net supply. Accordingly, the market price $\rho_m(s_m)$ is defined as

$$\rho_m(s_m) = \mu_m - \gamma_m s_m. \quad (6)$$

Note that the market price may become negative when the transaction amount exceeds a

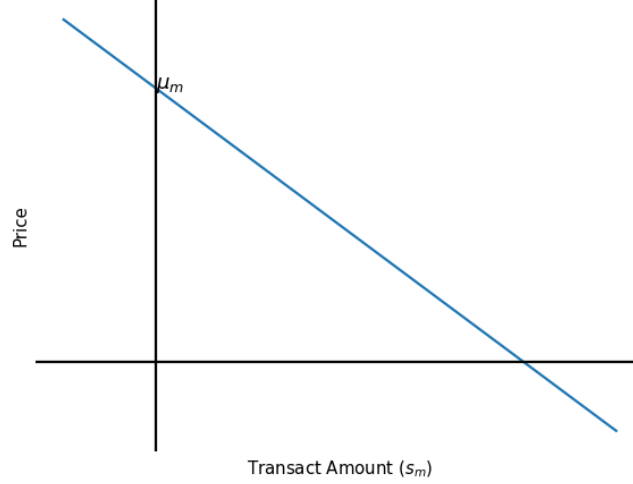


Figure 2: Inverse-demand curve.

threshold, i.e., $s_m > \mu_m/\gamma_m$. In this regime, DSs must effectively pay to export electricity. Conversely, when demand exceeds supply (i.e., $s_m < 0$), the market price increases above the baseline LMP.

We assume that the interconnection lines operate at high voltage levels; consequently, only active power flows are considered on these lines. These interconnections impact the active power balance constraints (3a) at the buses to which they are connected. The resulting active power balance constraint, for all $i \in \mathcal{B}_n$, is given by

$$\sum_{j \in \mathcal{G}_{n,i}} p_{n,j}^g - \sum_{j \in \mathcal{B}_{n,i}} p_{n,j}^d - \sum_{m: (i,m) \in \mathcal{I}_n} y_{nim} - p_{n,i}^f = 0. \quad (7)$$

Similarly, transactions with the TS influence the objective function (4). Let $\mathbf{y}_{-n} := \{y_k\}_{k \neq n}$

denote the decisions of all DSs except DS n . We define the market interaction term as

$$\begin{aligned}
g_n(y_n; \mathbf{y}_{-n}) &:= - \sum_{(i,m) \in \mathcal{I}_n} (\mu_m - \gamma_m s_m) y_{nim} \\
&= - \sum_{(i,m) \in \mathcal{I}_n} (\mu_m - \gamma_m \sum_{n' \in \mathcal{N}} \sum_{j: (j,m) \in \mathcal{I}_{n'}} y_{n'jm}) y_{nim} \\
&= - \underbrace{\sum_{(i,m) \in \mathcal{I}_n} \mu_m y_{nim}}_{(a)} \\
&\quad + \underbrace{\sum_{(i,m) \in \mathcal{I}_n} \gamma_m \left(\sum_{j: (j,m) \in \mathcal{I}_n} y_{njm} \right) y_{nim}}_{(b)} \\
&\quad + \underbrace{\sum_{(i,m) \in \mathcal{I}_n} \gamma_m \left(\sum_{n' \in \mathcal{N}, n' \neq n} \sum_{j: (j,m) \in \mathcal{I}_{n'}} y_{n'jm} \right) y_{nim}}_{(c)}. \tag{8}
\end{aligned}$$

Observe that terms (a) and (b) are linear and quadratic functions of the decision variables of the n -th DS. In contrast, term (c) contains bilinear terms involving decision variables of DSs that participate in the same market m . Consequently, the transaction function depends on the decisions of other DSs through term (c). Since $\gamma_m > 0$, the resulting transaction function is convex in the decision variables. Then, the updated objective function of DS n can be written as

$$f_n(y_n; \mathbf{y}_{-n}) = f_n(p_n^g) + g_n(y_n; \mathbf{y}_{-n}). \tag{9}$$

2.3 Mathematical Structure

We consolidate all the decision variables for the n -th DS into two vectors: the dispatch vector $x_n = ((p_{n,i}^g, q_{n,i}^g, p_{n,j}^d, p_{n,j}^f, q_{n,j}^f, c_{n,jj}, c_{n,ij}, s_{n,ij})_{i \in \mathcal{G}_n, j \in \mathcal{B}_n, (i,j) \in \mathcal{L}_n})$, and the transaction vector $y_n = ((y_{nim})_{(i,m) \in \mathcal{I}_n})$. The transaction variables y introduce coupling among DSs through shared markets. Let $y := (y_n)_{n \in \mathcal{N}}$ denote the collection of transaction decisions. For each $n \in \mathcal{N}$, $\mathbf{y}_{-n} := (y_{n'})_{n' \in \mathcal{N} \setminus \{n\}}$. Then, the decision problem of DS n is given by

$$\min_{y_n \in \mathcal{Y}_n} f_n(y_n; \mathbf{y}_{-n}). \tag{10}$$

where $f_n(y_n; \mathbf{y}_{-n})$ is given by (9) and the $\mathcal{Y}_n := \{y_n \text{ satisfying (1), (2), (3b), (7)}\}$. Note that the feasible region of each DS depends solely on its own parameters and decision variables, and is independent of the decisions of other DSs. Hence, the feasible region is private to each DS. The coupling among DSs arises exclusively through the objective function via the transaction decisions of other players, denoted by $\mathbf{y}_{-n}$.

Under this structure, the problem can be formulated as a standard Nash game [Nash, 1951]. Consequently, the collective strategy of all DSs corresponds to a Nash equilibrium, which we formally define next.

Definition 2.1. A tuple $\mathbf{y}^*$ is a Nash equilibrium if

$$f_n(y_n^*; \mathbf{y}_{-n}^*) \leq f_n(y_n; \mathbf{y}_{-n}^*) \quad \forall y_n \in \mathcal{Y}_n,$$

for all $n = 1, \dots, \mathcal{N}$.

Under this definition, no DS can unilaterally improve its objective by deviating from its equilibrium strategy y_n^* . Before presenting solution approaches for computing such equilibria in the next section, we extend the model to a stochastic setting in which electricity demand is uncertain.

In practice, electricity demand is subject to uncertainty due to load variability and forecasting errors in DSs. We model this uncertainty by a random vector ξ_n for each DS n , defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Accordingly, a two-stage stochastic Nash equilibrium problem [Xu and Zhang, 2013] is formulated as follows, where the first-stage decisions y_n determine market transactions, and the second-stage decisions x_n are adjusted after the realization of ξ_n .

$$\min_{y_n} g_n(y_n; \mathbf{y}_{-n}) + \mathbb{E}[Q(y_n; \xi_n)] \quad (11a)$$

where,

$$\begin{aligned} Q(y_n; \xi_n) = & \min_{x_n \in \mathcal{X}_n} h_n(x_n; \xi_n) \\ \text{s.t.} \quad & \sum_{j \in \mathcal{G}_{n,i}} p_{n,j}^g - \sum_{m: (i,m) \in \mathcal{I}_n} y_{nim} \\ & - p_{n,i}^f = \sum_{j \in \mathcal{B}_{n,i}} p_{n,j}^d(\xi_n). \end{aligned} \quad (11b)$$

Here $g_n(y_n; \mathbf{y}_{-n})$ is defined in (8), $h_n(x_n; \xi_n)$ denotes the system cost (4) and $\mathcal{X}_n := \{x_n \text{ satisfying (1), (2), (3b)}\}$. Here, the stochasticity comes from the active electricity demand $p_{n,i}^d$ at every bus $i \in \mathcal{B}_n$. In general, $\mathbb{E}[Q(y_n; \xi_n)]$ cannot be evaluated in a closed form and must be approximated via discretization.

We approximate it using the sample average approximation (SAA) method. Let $\{\xi_n^s\}_{s=1}^S$ be a set of independent and identically distributed (i.i.d.) samples drawn from the distribution of the random vector ξ_n . The expectation is then approximated by the empirical average over these samples.

Under the SAA scheme, the two-stage stochastic Nash equilibrium problem (11) is approximated by replacing the expectation with its sample average, yielding the following problem for each $n \in \mathcal{N}$:

$$\begin{aligned} \min_{y_n, \{x_n^s\}_{s=1}^S} & g_n(y_n; \mathbf{y}_{-n}) + \frac{1}{S} \sum_{s=1}^S h_n(x_n^s; \xi_n^s) \\ \text{s.t.} \quad & \sum_{j \in \mathcal{G}_{n,i}} p_{n,j}^{g,s} - \sum_{m: (i,m) \in \mathcal{I}_n} y_{nim} - p_{n,i}^{f,s} \\ & = \sum_{j \in \mathcal{B}_{n,i}} p_{n,j}^d(\xi_n^s), \quad x_n^s \in \mathcal{X}_n, s = 1, 2, \dots, S. \end{aligned} \quad (12)$$

In this formulation, each scenario $s = 1, \dots, S$ corresponds to an independent realization of the uncertain demand. As S increases, the SAA problem provides a consistent approximation of the original stochastic problem, and its optimal solution converges to that of the true problem under suitable regularity conditions. In the next section, we present solution approaches for both the deterministic and stochastic models.

3 Proximal BR Algorithms

In this section, we present solution algorithms for the Nash equilibrium problem (10) and two-stage stochastic Nash equilibrium problem (11). In these algorithms, each DS updates its

strategy independently, given the decisions of other DSs. Such methods are commonly referred to as best response (BR) schemes [Basar and Olsder, 1999]. However, BR schemes do not, in general, guarantee convergence to a Nash equilibrium. An example illustrating this limitation can be found in [Facchinei et al., 2011].

To address this issue, it has been proposed to incorporate a proximal term into the objective function [Facchinei et al., 2011, Facchinei and Pang, 2010]. The resulting proximal BR scheme has been shown to generate a sequence of iterates that converges to a Nash equilibrium when each DS-specific optimization problem is convex. Motivated by these results, we adopt a proximal objective function of the form:

$$\min f_n(y_n; \mathbf{y}_{-n}^k) + \frac{\mu}{2} \|y_n - y_n^{k-1}\|^2, \quad (13)$$

where k denotes the iteration index. Let y_n^k denote the optimal solution at iteration k . The proximal parameter $\mu > 0$, which is an input to the algorithm, controls the deviation from the previous iterate y_n^{k-1} .

Proximal BR algorithm is also an effective method for solving two-stage stochastic Nash equilibrium problems. At iteration k , given the strategies of other DSs, each player $n \in \mathcal{N}$ solves a regularized problem of the form

$$\min_{y_n, \{x_n^s \in \mathcal{X}_n\}_{s=1}^S} g_n(y_n; \mathbf{y}_{-n}^k) + \frac{1}{S} \sum_{s=1}^S h_n(x_n^s; \xi_n^s) + \frac{\mu}{2} \|y_n - y_n^{k-1}\|^2, \quad (14)$$

It is shown in [Xu and Zhang, 2013] that, under mild conditions, a statistical estimator (i.e., a Nash equilibrium or a Nash C-stationary point) obtained from the sample average approximation of the equilibrium problem converges almost surely to its true counterpart.

The performance of a BR scheme is primarily influenced by two factors: the order in which the DS-specific problems are solved and the timing of updates to the solution vector $\mathbf{y}$. The selection of DSs can follow either a cyclic or a randomized order. Similarly, the decision vector $\mathbf{y}$ can be updated in either an immediate or a coordinated manner. Based on these two components, we assess three variants of the proximal BR algorithm for our coordinated T-D system problem.

3.1 Cyclic Order Coordinated Updates

The first algorithm employs a cyclic ordering of the DS problems combined with coordinated updates. Specifically, the DS problems are solved in the same fixed order at each iteration. Under the coordinated update scheme, the solution vector $\mathbf{y}^k$ and the market prices are updated only after all DS problems have been solved. This algorithm is presented in Algorithm 1.

The algorithm takes all DSs, markets, and interconnection information as inputs. Between lines 3 and 4, we setup the objective function, solve the optimization problems separately and store the solution in their respective vector y_n^k for all individual DS problems. Notice that prices at markets connected to a specific DS are used in updating its objective function. After solving all the DS problems, in line 5, we update the market information using the consolidated decision vector $\mathbf{y}^k$, which affects the objective functions of DS models. We run the algorithm for a prespecified termination condition. It is worthwhile to note that the price updates are carried out once per iteration.

Algorithm 1 Cyclic order coordinated updates

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- 1: **Input:** DS $\mathcal{N} = \{1, 2, \dots, N\}$, Markets $\mathcal{M} = \{1, 2, \dots, M\}$, Interconnections $(\{\mathcal{I}_n\}_{n \in \mathcal{N}})$, $\mu > 0$, $k \leftarrow 0$.
 - 2: **while** Stopping criteria not met **do**
 - 3: **for** $n = 1$ to N **do**
 Setup objective for n^{th} DS using $\rho_m(s_m^{k-1})$, $\forall m \in \mathcal{M}$ as shown in (13). Solve and extract solutions

$$y_n^k \in \arg \min f_n(y_n; \mathbf{y}_{-n}^{k-1}) + \frac{\mu}{2} \|y_n - y_n^{k-1}\|^2.$$
 - 4: **end for**
 - 5: Using $\mathbf{y}^k = (y_1^k, y_2^k, \dots, y_N^k)$, update the prices at all markets as:

$$\rho_m(s_m^k) = \mu_m - \gamma_m s_m^k \quad \forall m \in \mathcal{M}.$$
 - 6: $k \leftarrow k + 1$.
 - 6: **end while**
-

By replacing the objective function in line 3 with (14), we obtain a stochastic variant of Algorithm 1 for solving the extensive form (12). In this variant, each DS solves a scenario-based optimization problem at every iteration, where the expectation term is approximated using a finite set of sampled realizations. Specifically, for each DS $n \in \mathcal{N}$, the optimization problem incorporates the SAA of the second-stage cost, while retaining the proximal regularization term to ensure stability of the updates. The resulting algorithm preserves the same cyclic ordering and coordinated update structure as in Algorithm 1.

As a result, the sequence of iterates $\{\mathbf{y}^k\}$ is generated by repeatedly solving SAA-based subproblems and updating the market prices accordingly. Under standard convexity assumptions, this stochastic proximal BR scheme is expected to converge to a Nash equilibrium as the number of samples increases.

3.2 Cyclic Order Immediate Updates

The second algorithm also adopts a cyclic ordering of the DS problems. However, unlike the coordinated update scheme, market prices are updated immediately after solving each individual DS problem. The detailed procedure is presented in Algorithm 2.

Similar to algorithm 1, the second algorithm takes the same inputs and runs until a prespecified termination condition is met. Notice that the algorithm updates the consolidated decision vector $\mathbf{y}_n^{k,n}$ vector before solving every individual DS problem in line 3. We capture this distinction by using an addition superscript in the notation of the consolidated decisions vector ($\mathbf{y}_n^{k,n}$ as opposed to $\mathbf{y}_n^k$ in Algorithm 1). This change results in multiple (N times in each iteration) updates to the market prices. Consequently, the objective function of each DS model reflects the latest price information. Then, each DS model solves for their optimal solutions.

Replacing the objective function with (14) yields the stochastic version of Algorithm 2, where each DS solves an SAA problem with immediate price updates.

Algorithm 2 Cyclic order immediate updates

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- 1: **Input:** DS $\mathcal{N} = \{1, 2, \dots, N\}$, Markets $\mathcal{M} = \{1, 2, \dots, M\}$, Interconnections $(\{\mathcal{I}_n\}_{n \in \mathcal{N}})$, $\mu > 0$, $k \leftarrow 0$.
 - 2: **while** Stopping criteria not met **do**
 - 3: **for** $n = 1$ to N **do**

Using $\mathbf{y}_{-n}^{k,n} = (y_1^k, y_2^k, \dots, y_{n-1}^k, y_{n+1}^{k-1}, \dots, y_N^{k-1})$, update the prices at all markets as:

$$\rho_m(s_m^{k,n}) = \mu_m - \gamma_m s_m^{k,n} \quad \forall m \in \{M\}.$$

Setup objective for n^{th} DS using $\rho_m(s_m^{k,n})$ as shown in (13). Solve and extract solutions

$$y_n^k \in \arg \min f_n(y_n; \mathbf{y}_{-n}^k) + \frac{\mu}{2} \|y_n - y_n^{k-1}\|^2.$$

- 4: **end for**
 $k \leftarrow k + 1$.
 - 5: **end while**
-

3.3 Random Order Immediate Updates

The final algorithm follows a similar structure to the previous one, with the key difference that the order of solving the DS problems is selected randomly at each iteration. Under a coordinated update scheme, the ordering of DS updates does not affect the outcome; thus, cyclic and random orderings yield equivalent algorithms. Consequently, when random ordering is employed, we focus only on the immediate update scheme. The resulting procedure is presented in Algorithm 3.

Algorithm 3 Random order immediate updates

-
- 1: **Input:** DS $\mathcal{N} = \{1, 2, \dots, N\}$, Markets $\mathcal{M} = \{1, 2, \dots, M\}$, Interconnections $(\{\mathcal{I}_n\}_{n \in \mathcal{N}})$, $\mu > 0$, $k \leftarrow 0$.
 - 2: **while** Stopping criteria not met **do**
 Generate a random permutation, $\mathcal{P} = (p_1, p_2, \dots, p_N)$.
 - 3: **for** $n = 1$ to N **do**
 - 4: Using $\mathbf{y}_{-n}^{k,n} = (y_{j:p_j < n}^k, y_{j:p_j > n}^{k-1})$, update the prices at all markets as:

$$\rho_m(s_m^{k,n}) = \mu_m - \gamma_m s_m^{k,n} \quad \forall m \in \mathcal{M}.$$

Setup objective for n^{th} DS using $\rho_m(s_m^{k,n})$ as shown in (13). Solve and extract solutions

$$y_n^k \in \arg \min f_n(y_n; \mathbf{y}_{-n}^k) + \frac{\mu}{2} \|y_n - y_n^{k-1}\|^2.$$

- 5: **end for**
 $k \leftarrow k + 1$
 - 6: **end while**
-

The difference between algorithms 2 and 3 is in line 2. That is, in algorithm 3 we pick a random permutation to determine the order of solving the DS optimization problems. At the beginning of the n^{th} round, we have solved optimization problems for DSs indexed by $p_1, p_2, \dots, p_n$. Therefore, we have their optimal solutions $y_{j:p_j < n}^k$ identified in iteration k . However, we only have the optimal solutions from the previous iteration for DSs indexed by $p_{n+1}, \dots, p_N$, that is, $y_{j:p_j > n}^{k-1}$. The resulting consolidated vector $\mathbf{y}_{-n}^{k,n}$ is used in computing the market prices in line 4.

Table 1: Performance of Algorithms 1–3 in the three-DS and six-DS settings.

Setting	μ	Algorithm 1: CC			Algorithm 2: CI			Algorithm 3: RI		
		Iters.	Objective	Time	Iters.	Objective	Time	Iters.	Objective	Time
Three-DS	0	24	982.20	22.74	13	982.20	17.30	18	982.20	19.70
	0.125	24	982.20	22.04	13	982.20	17.07	18	982.20	18.67
	1	24	982.20	22.54	13	982.20	18.26	19	982.20	20.87
	8	22	982.20	22.24	14	982.20	16.85	19	982.20	19.49
	64	31	982.20	29.12	25	982.20	22.99	28	982.20	24.23
Six-DS	0	100+	–	–	19	8659.39	28.72	22	8659.39	32.66
	0.125	100+	–	–	19	8659.39	28.23	22	8659.39	31.57
	1	100+	–	–	19	8659.39	32.09	22	8659.39	32.28
	8	100+	–	–	19	8659.39	30.53	22	8659.39	30.85
	64	100+	–	–	25	8659.39	36.01	31	8659.39	42.31

The rest of the steps are exactly similar to the case of cyclical order immediate updates. Similarly, when taking the objective function as (14), we obtain the stochastic version of Algorithm 3.

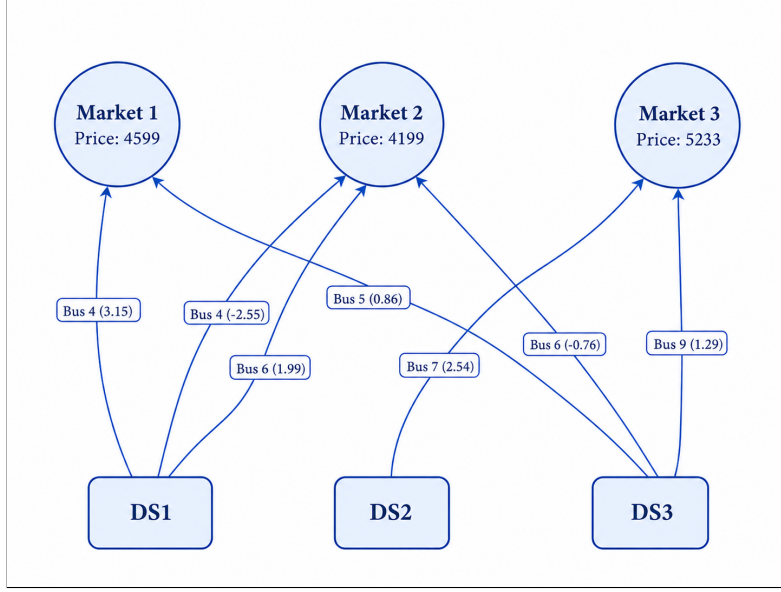
4 Experiments

In this section, we report the computational results for the three variants of the proximal BR algorithm and its stochastic counterpart introduced in Section 3. We first describe the case study setup and then present the numerical results.

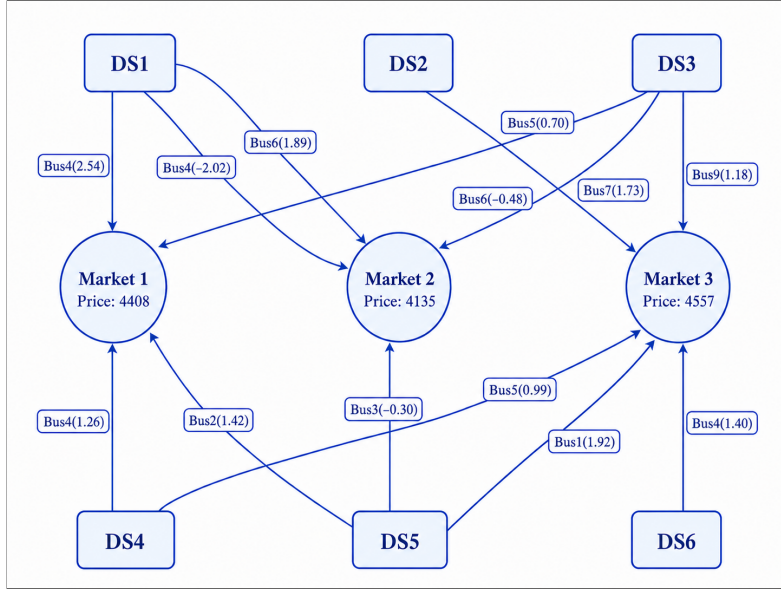
4.1 Deterministic Problem Settings and Results

For the numerical experiments, we consider two market settings. The first setting consists of three markets and three DSs, while the second setting consists of three markets and six DSs. The DSs are constructed by modifying the IEEE 9-bus, IEEE 14-bus, and IEEE 30-bus test systems [Christie,]. We assume that there are no direct physical connections among DSs; hence, interactions among DSs occur only through the market prices. Moreover, the baseline market prices μ_m in the inverse demand function are set to \$5000, \$4000, and \$6000 for Markets 1, 2, and 3, respectively. The corresponding price-sensitivity parameters γ_m are set to 100, 150, and 200. All algorithms were implemented in Julia 1.12. The SOCP relaxation models were constructed using PowerModels v0.21.5, and the resulting optimization problems were solved using Ipopt v1.12.1. All experiments were conducted on a computer with an Intel Core i7 2.20 GHz processor and 32 GB of RAM.

We evaluate the three proximal BR algorithms under the two deterministic market settings and different values of the proximal parameter μ . The results for the three-DS and six-DS settings are reported in Table 1. It shows that all convergent cases reach the same objective value for each market setting. This observation suggests that the proximal term primarily affects the convergence trajectory rather than the final equilibrium objective. Therefore, the comparison among the three algorithms is mainly based on convergence speed, measured by the number of iterations and computational time.



(a) Three-DS setting



(b) Six-DS setting

Figure 3: Market prices for the three-DS and six-DS settings.

Among the three algorithms, the cyclic-order immediate-update scheme generally achieves the fastest convergence. For the three-DS setting, all three algorithms converge for all tested values of μ , while Algorithm 2 requires the fewest iterations in most cases. For the six-DS setting, Algorithm 1 does not converge within 100 iterations, whereas Algorithms 2 and 3 converge to the same objective value. This result indicates that immediate price updates are more effective than coordinated updates in the proposed market setting. In addition, increasing the proximal parameter generally slows the update process, especially for Algorithm 3 in the six-DS case.

Fig. 3 illustrates the interconnection structure, final market prices, and interconnection decisions obtained by Algorithm 2 with proximal parameter $\mu = 2$ for both the three-DS and six-DS settings. The line labels specify the connected DS bus and the converged transaction amount (positive values indicate sales to the market, whereas negative values indicate purchases from the market). For example, in the three-DS setting, DS 1 is connected to Market 2 through

both Bus 4 and Bus 6, showing that the same DS can interact with the same market through multiple buses.

Compared with the three-DS case, the six-DS case yields lower prices across all three markets, since additional DSs participate in the coordination system and provide more generation resources. This suggests that broader DS participation can improve market competitiveness and help reduce market prices.

In the six-DS setting, the market prices converge to \$4408, \$4135, and \$4557 for Markets 1, 2, and 3, respectively, from the baseline prices of \$5000, \$4000, and \$6000. Although the price at Market 2 increases slightly, the initially large price gap across markets is substantially reduced, mainly due to the decrease in the initially high price at Market 3. This indicates that coordination through shared markets can mitigate price disparities even when the DSs are not directly connected.

We next examine the convergence behavior of the interconnection decisions. In the three-DS setting, there are seven interconnections between DSs and markets. Fig. 4 shows the convergence of these interconnection decisions using Algorithm 2 with $\mu = 2$. One interconnection from DS 1 to Market 2 converges to a negative value, which indicates that DS 1 purchases electricity from Market 2. Meanwhile, another interconnection from DS 1 to Market 2 converges to a positive value, indicating that DS 1 sells electricity to the same market. This demonstrates that a DS with multiple interconnections can simultaneously buy from and sell to markets, depending on the local network conditions and market prices.

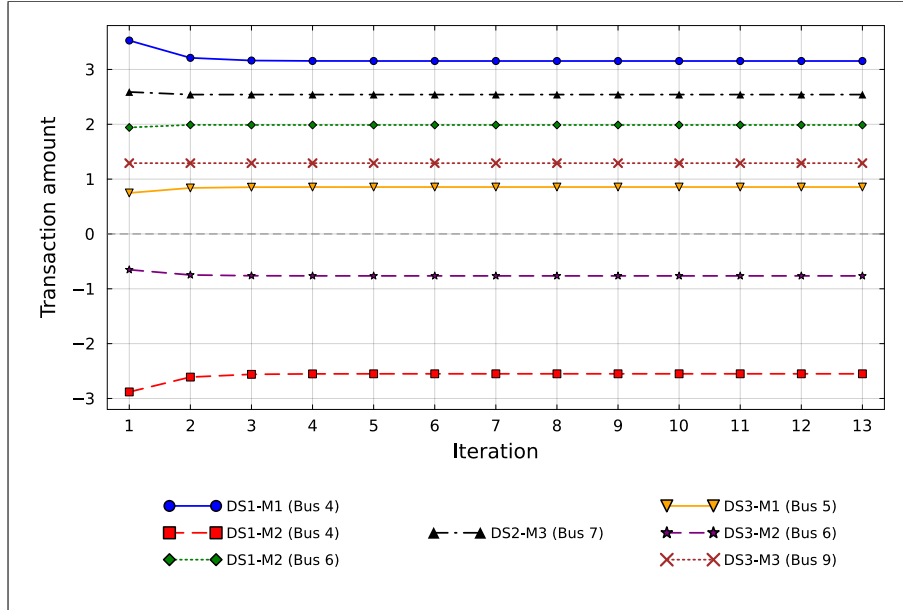


Figure 4: Convergence of Interconnections.

A similar pattern is observed for DS 3. This behavior can be interpreted as a form of market arbitrage: a DS may purchase electricity from a relatively low-price market and sell electricity to a relatively high-price market. However, such arbitrage is naturally limited by the inverse demand curve. As a DS purchases electricity from a market, the market price increases; as it sells electricity to a market, the market price decreases. Therefore, the price adjustment mechanism prevents unlimited arbitrage while still allowing DSs to benefit from flexible market participation.

Table 2: Performance of pBR+SAA with 10 samples under different $\hat{\sigma}$ and μ for the three-DS and six-DS settings. Time is measured in seconds.

$\hat{\sigma}$	μ	Three-DS Setting									Six-DS Setting					
		CC			CI			RI			CI			RI		
		Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time
0.05	0	21	1262.68	22.89	12	1262.68	17.72	13	1262.68	18.64	17	9120.66	32.04	19	9120.66	33.35
	0.125	21	1262.68	23.25	12	1262.68	18.04	13	1262.68	18.67	17	9120.66	31.17	19	9120.66	35.79
	1	21	1262.68	24.49	12	1262.68	19.05	14	1262.68	20.41	17	9120.66	31.40	19	9120.66	34.03
	4	21	1262.68	22.59	12	1262.68	18.18	15	1262.68	19.56	15	9120.66	29.28	20	9120.66	35.16
	16	20	1262.68	22.13	14	1262.68	19.65	16	1262.68	20.14	18	9120.66	33.45	21	9120.66	36.83
	32	22	1262.68	24.58	16	1262.68	20.80	19	1262.68	22.16	19	9120.66	34.01	23	9120.66	39.68
0.10	0	21	1330.92	22.83	12	1330.92	16.90	14	1330.92	18.53	17	9597.83	31.02	19	9597.83	34.42
	0.125	21	1330.92	23.22	12	1330.92	17.19	14	1330.92	18.95	17	9597.83	32.19	19	9597.83	34.19
	1	21	1330.92	22.71	12	1330.92	17.01	14	1330.92	18.28	17	9597.83	32.16	19	9597.83	33.80
	4	21	1330.92	22.48	12	1330.92	17.25	14	1330.92	19.99	15	9597.83	30.53	19	9597.83	35.13
	16	22	1330.92	25.88	14	1330.92	18.38	15	1330.92	20.71	18	9597.83	32.81	20	9597.83	36.69
	32	24	1330.92	25.94	16	1330.92	19.97	18	1330.92	20.64	18	9597.83	33.65	23	9597.83	40.42
0.15	0	24	1434.47	26.28	11	1434.47	16.93	15	1434.47	19.99	17	10079.97	34.89	18	10079.97	36.16
	0.125	24	1434.47	25.96	11	1434.47	17.04	15	1434.47	19.88	17	10079.97	32.42	18	10079.97	33.30
	1	22	1434.47	24.92	12	1434.47	18.29	15	1434.47	20.07	18	10079.97	37.61	18	10079.97	33.63
	4	22	1434.47	23.76	12	1434.47	17.71	15	1434.47	19.42	15	10079.97	34.04	18	10079.97	33.48
	16	20	1434.47	22.00	13	1434.47	18.78	17	1434.47	20.83	18	10079.97	35.14	20	10079.97	37.12
	32	24	1434.47	25.15	16	1434.47	19.99	19	1434.47	22.27	18	10079.97	35.93	22	10079.97	39.36
0.20	0	21	2183.87	25.91	12	2183.87	19.07	13	2183.87	19.38	17	10561.28	34.34	19	10561.28	35.63
	0.125	21	2183.87	24.47	12	2183.87	17.89	13	2183.87	19.06	17	10561.28	36.53	19	10561.28	36.02
	1	21	2183.87	26.66	12	2183.87	19.28	14	2183.87	19.65	17	10561.28	35.29	19	10561.28	37.67
	4	21	2183.87	27.24	12	2183.87	18.56	15	2183.87	20.75	16	10561.28	34.26	19	10561.28	37.92
	16	22	2183.87	27.90	14	2183.87	19.79	17	2183.87	21.61	17	10561.28	34.79	22	10561.28	49.09
	32	22	2183.87	25.62	16	2183.87	20.84	19	2183.87	22.60	19	10561.28	35.36	22	10561.28	41.29

4.2 Stochastic Problem Settings and Results

We next consider the stochastic setting, where the active electricity demand p_n^d is uncertain. The demand is sampled from a normal distribution with mean μ_d and standard deviation $\sigma_d = \mu_d \hat{\sigma}$, where μ_d is the deterministic demand level and $\hat{\sigma} \in (0, 1)$ controls the uncertainty level. All other parameters are the same as in the deterministic experiments. The pBR+SAA algorithms are tested under different uncertainty levels, proximal parameters, and SAA sample sizes for both the three-DS and six-DS settings.

Table 2 reports the performance of pBR+SAA with 10 samples for both the three-DS and six-DS settings. Consistent with the deterministic case, the coordinated-update (CC) scheme failed to converge within the maximum of 100 iterations for all six-DS instances. Therefore, no objective value or computational time is reported for CC in the six-DS results of Table 2. For each fixed setting and uncertainty level, all convergent algorithms reach the same objective value, indicating that the update rule mainly affects convergence speed rather than the final equilibrium objective. In the three-DS setting, all three algorithms converge for all tested values of $\hat{\sigma}$ and μ , and Algorithm 2 generally requires the fewest iterations and the shortest computational time. In the six-DS setting, the coordinated-update scheme fails to converge within 100 iterations in all tested cases, while both immediate-update schemes converge reliably. Algorithm 2 again requires fewer iterations than Algorithm 3 in most cases, showing the advantage of immediate price updates in larger systems.

The results also show that higher demand uncertainty leads to higher objective values. For example, with 10 samples in the six-DS setting, the objective value increases from 9120.66 at

Table 3: Performance of pBR+SAA with $\mu = 4$ under different sample sizes and $\hat{\sigma}$ for the three-DS and six-DS settings. Time is measured in seconds.

$\hat{\sigma}$	Sample size	Three-DS Setting									Six-DS Setting					
		CC			CI			RI			CI			RI		
		Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time	Iter.	Obj.	Time
0.05	10	21	1262.68	27.94	12	1262.68	17.46	14	1262.68	19.76	15	9120.66	26.99	20	9120.66	31.77
	50	21	1316.71	87.37	12	1316.71	50.99	16	1316.71	63.59	18	9277.41	122.19	22	9277.41	142.01
	100	22	1163.85	158.70	12	1163.85	95.20	15	1163.85	110.99	17	9130.86	209.61	19	9130.86	239.04
	250	22	1364.74	546.74	12	1364.74	303.93	14	1364.74	352.72	17	9177.10	726.19	20	9177.10	870.04
	500	21	1307.87	1099.74	12	1307.87	650.03	17	1307.87	917.12	17	9187.13	1536.56	20	9187.13	1826.73
0.10	10	22	1558.60	27.07	12	1558.60	19.87	14	1558.60	20.03	15	9597.83	30.53	19	9597.83	35.08
	50	22	1666.53	83.93	11	1666.53	49.38	15	1666.53	64.85	15	9921.34	107.38	22	9921.34	226.53
	100	21	1361.22	158.61	11	1361.22	86.69	14	1361.22	107.04	15	9636.57	197.81	20	9636.57	264.43
	250	21	1766.45	537.74	12	1766.45	313.39	14	1766.45	354.73	15	9731.39	944.35	21	9731.39	1389.71
	500	22	1652.48	1207.32	13	1652.48	734.89	18	1652.48	1012.12	16	9752.14	2074.61	20	9752.14	1957.30
0.15	10	21	1863.79	23.60	12	1863.79	17.27	13	1863.79	17.58	15	10079.97	28.59	18	10079.97	32.69
	50	26	2037.31	99.32	12	2037.31	51.92	15	2037.31	60.98	18	10584.98	124.88	22	10584.98	148.45
	100	21	1568.59	162.09	11	1568.59	85.77	14	1568.59	109.43	17	10165.80	226.36	18	10165.80	258.42
	250	24	2183.13	625.95	13	2183.13	326.01	14	2183.13	365.50	15	10322.26	722.88	20	10322.26	952.57
	500	24	2012.75	1261.98	12	2012.75	648.81	17	2012.75	916.44	16	10343.76	1651.48	20	10343.76	1982.17
0.20	10	21	2183.87	24.18	12	2183.87	19.68	14	2183.87	19.79	16	10561.28	31.78	19	10561.28	34.22
	50	17	2443.58	69.86	10	2443.58	46.99	12	2443.58	53.65	17	11200.87	121.90	22	11200.87	147.61
	100	22	1791.84	174.95	12	1791.84	106.43	15	1791.84	126.16	16	10663.16	229.87	18	10663.16	242.15
	250	21	1731.31	523.85	12	1731.31	309.22	14	1731.31	360.44	17	10804.54	1401.73	20	10804.54	1243.03
	500	21	1776.80	1165.19	12	1776.80	677.94	18	1776.80	992.76	17	10930.04	2651.92	19	10930.04	1897.36

$\hat{\sigma} = 0.05$ to 10561.28 at $\hat{\sigma} = 0.20$. This reflects the additional operating cost caused by higher demand variability.

Table 3 studies the effect of the SAA sample size with $\mu = 4$ for both settings. The number of pBR iterations remains relatively stable as the sample size increases, but the computational time grows substantially because each DS subproblem contains more scenarios. For example, in the three-DS setting with $\hat{\sigma} = 0.05$, the runtime of Algorithm 2 increases from 17.46 seconds with 10 samples to 650.03 seconds with 500 samples. These results highlight the trade-off between solution accuracy and computational effort in the SAA-based stochastic pBR framework.

To evaluate the out-of-sample performance of the deterministic and SAA solutions, we use 500 independently generated validation scenarios. The first-stage transaction decisions are obtained from SAA models trained with $\hat{\sigma} = 0.20$ and then tested under validation scenarios with different uncertainty levels. For each candidate first-stage solution y , we fix the transaction decisions and solve the corresponding second-stage problem for each validation scenario. The out-of-sample objective value is estimated by

$$\hat{z}(y) = g(y) + \frac{1}{M} \sum_{j=1}^M Q(y, \tilde{\xi}_j),$$

where $\tilde{\xi}_1, \dots, \tilde{\xi}_M$ are the validation scenarios. We report the sample mean, standard deviation, confidence interval, and the number of infeasible validation instances.

Table 4 summarizes the validation results under different SAA sample sizes and validation uncertainty levels. The deterministic expected-value solution is not robust to demand uncertainty, as it becomes infeasible for all validation scenarios in the reported case. In contrast, the SAA solutions provide better out-of-sample feasibility under moderate uncertainty. For example, when $\hat{\sigma} = 0.05$, all SAA solutions are feasible across the 500 validation scenarios. When $\hat{\sigma} = 0.15$, increasing the SAA sample size reduces the number of infeasible cases from 357/500 with 100 samples to 22/500 with 500 samples. However, when the validation uncertainty level is high, such as $\hat{\sigma} = 0.20$, infeasibility remains significant even with larger sample sizes.

Overall, the stochastic results show that the proximal BR variants mainly differ in convergence efficiency rather than the final equilibrium objective. Immediate-update schemes outperform the coordinated-update scheme, especially in the larger six-DS setting. Higher uncertainty increases the objective value, while larger SAA sample sizes mainly increase runtime rather than the number of pBR iterations. In addition, the validation results suggest that SAA improves the robustness of first-stage decisions relative to the deterministic expected-value model, but its performance depends on both the validation uncertainty level and the number of training scenarios.

Table 4: Out-of-sample validation results under different SAA sample sizes.

Setting	Mean	Std. Dev.	Confidence Interval	Infeasible
Deterministic				
$\hat{\sigma} = 0.05$	–	–	–	500/500
SAA with sample size = 100				
$\hat{\sigma} = 0.05$	1826.37	20.38	[1824.59, 1828.16]	0/500
$\hat{\sigma} = 0.10$	1836.53	40.93	[1832.93, 1840.14]	4/500
$\hat{\sigma} = 0.15$	1859.70	62.27	[1849.49, 1869.90]	357/500
$\hat{\sigma} = 0.20$	–	–	–	500/500
SAA with sample size = 250				
$\hat{\sigma} = 0.05$	2211.86	20.35	[2210.08, 2213.65]	0/500
$\hat{\sigma} = 0.10$	2222.15	40.74	[2218.58, 2225.73]	0/500
$\hat{\sigma} = 0.15$	2243.98	61.69	[2238.03, 2249.93]	87/500
$\hat{\sigma} = 0.20$	2265.71	68.19	[2240.89, 2290.52]	471/500
SAA with sample size = 500				
$\hat{\sigma} = 0.05$	2168.61	20.36	[2166.83, 2170.40]	0/500
$\hat{\sigma} = 0.10$	2178.75	40.76	[2175.18, 2182.33]	0/500
$\hat{\sigma} = 0.15$	2198.77	61.09	[2193.29, 2204.25]	22/500
$\hat{\sigma} = 0.20$	2229.25	81.40	[2215.47, 2243.04]	366/500

5 Conclusions

In this paper, we proposed a multiagent coordination framework for TS and DS operations. Multiple DSs interact with the TS through shared market nodes, and market prices are updated using an inverse demand function. This structure allows DSs to make independent decisions while coordinating through limited information exchange.

We formulated the problem as a noncooperative Nash game and developed three proximal best-response algorithms. The numerical results show that all convergent algorithms reach the same equilibrium objective, but differ in convergence speed. Among them, the cyclic-order immediate-update scheme performs the best in terms of iteration count and computational time. The results also show that immediate price updates become especially important in the larger six-DS setting, where the coordinated-update scheme fails to converge within the iteration limit.

The deterministic experiments indicate that broader DS participation can reduce price disparities across markets. The stochastic experiments further show that higher demand uncertainty increases objective values, while larger SAA sample sizes increase computational time.

Out-of-sample validation results suggest that the SAA approach provides more robust first-stage transaction decisions than the deterministic expected-value model, although high uncertainty can still lead to infeasibility.

While we focused on a setting involving interactions between transmission and DSs, we can view it as a special case of a more general setting with multiple agents that interact with each other in a spatially separated market. Examples include regional trading of agricultural goods, retail markets, and multiagent resource allocation using market-based mechanisms. These application areas bring forth specific modeling and algorithmic challenges that we will explore in our future research. The numerical experiments validate our approach of using a best-response procedure to handle the optimization problem. However, handling a greater number of agents or the case where the optimization problems encountered by individual agents are large-scale with a large number of scenarios will require algorithmic innovations. Adaptation of classical stochastic programming algorithms like the L-shaped or progressive hedging algorithms provides a fruitful avenue to explore. The challenge in a multiagent setting is to guarantee convergence of these algorithms and the nature of solutions that we obtain. We will undertake the tasks of improving the scalability of the stochastic algorithms and developing more robust formulations for high-demand uncertainty as part of our future work.

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